EDUVERSE

Developed By:

Shreyansh Dhanani - U02001625

Pratham Modi - U01994470

Bhargav Kakadiya- U01996543

Jay Kakadiya – U01987717

Guided By:

Kshitij Sharma

Institution:

Pace University, Seidenberg School of Computer Science & Information Systems

AGENDA

1. Team Member Roles and Responsibilities
2. Improvements Made from Professor Feedback
3. Project Description
4. Team working agreement
5. Personas
6. Minimum Viable Product (MVP)
7. Technologies
8. Algorithms
9. Diagrams
10. Product Backlog
11. Sprint 1 Backlog
12. Metrics (Velocity, Burndown Charts, Completed/Committed Ratio)
13. Retrospective
14. Sprint 2 Planned Stories
15. Project Demo - Sprint 1
16. GitHub Link
17. Live Application Demo

1. TEAM MEMBER ROLES AND RESPONSIBILITIES

Team Member	Role	Responsibilities
Shreyansh Dhanani	Full Stack Developer / Tech Lead	- Backend architecture design (NestJS, ExpressJS) - Database schema design and management - Code review and technical documentation
Pratham Modi	Backend Developer	- API development and integration - System deployment and performance optimization - Code review and technical documentation
Bhargav Kakadiya	Frontend Developer	- Frontend development (ReactJS) - User experience testing - Frontend documentation
Jay Kakadiya	UI/UX Designer	- UI/UX design and implementation - Component architecture

2.IMPROVEMENTS MADE FROM PROFESSOR FEEDBACK

1. Personas :- Updated based on professor's feedback: personas now highlight user frustrations and motivations without suggesting platform solutions."

2. Retrospective:- "Based on professor's Sprint 0 feedback, we focused on making retrospective more specific to actual sprint work, setting measurable goals tied to backlog.

3. PROJECT DESCRIPTION

- The goal of the comprehensive online learning platform Eduverse is to provide a smooth educational ecosystem that links universities, super administrators, course suppliers, and students.
- Motivated by websites such as Coursera and Udemy, Eduverse facilitates communication between students and academic institutions by allowing:
- Universities could buy subscription plans and give their students free access to courses.
- Course providers to produce, oversee, and profit from instructional materials
- Students can obtain certifications and take advantage of a variety of learning opportunities.
- Super administrators to supervise user management, analytics, and platform operations
- By providing an integrated solution with role-based access, institutional communication capabilities, and centralized administrative control, the platform seeks to address the disjointed experience of current solutions.

4. TEAM WORKING AGREEMENT

Meeting Schedule:

- **Daily Stand-up:** 30 minutes via video/voice call for daily updates and task discussion (10:00 AM, Monday–Friday)
- **Sprint Planning:** Every Monday, 45 minutes to 1 hour
- **Sprint Review/Retrospective:** Every Friday, 45 minutes
- **Ad-hoc Meetings:** Scheduled as needed via mutual agreement

Task Commitment & Deadlines:

- All tasks assigned during sprint planning must be updated on the project board within 24 hours
- Tasks should be updated with status (To Do, In Progress, Blocked, Done) daily
- If a deadline cannot be met, communicate at least 24 hours in advance

Work Distribution

- Work is distributed based on role responsibilities and individual capacity
- All members commit to equal contribution and shared ownership
- Code reviews are mandatory before merging to main branch
- Documentation is part of "done" for every task

Communication Protocols

- **Primary Channel:** WhatsApp group (urgent matters)
- **Secondary Channel:** GitHub Issues (task-specific discussions)
- **Response Time:** Within 4 hours during working hours (10:00 AM – 7:00 PM)
- **File Sharing:** Google Drive for documents, GitHub for code
- All major decisions must be documented in writing

5. PERSONAS

1. The Student – Maya Chen

Role: Final-year Computer Science Student / Aspiring Full-Stack Developer
Demographics: 21 years old, tech enthusiast, active learner

Personal Motivations:

- Wants to feel job-ready and confident in modern tech skills like React and Vue.js
- Dreams of building impressive projects that stand out to employers
- Prefers learning at her own pace, fitting study around her busy schedule

Frustrations:

- Her university classes feel stuck in the past—she's learning theory that won't help her land a job
- Every time she looks for online courses, she's overwhelmed by options and unsure which ones are actually good
- Worries that her self-learning won't be taken seriously by employers without some kind of proof

2. The University Administrator – Dr. Anya Sharma

Role: University Administrator

Demographics: 45 years old, PhD in Education, tech-savvy, budget-conscious

Personal Motivations:

- Wants to modernize the university's curriculum quickly to keep students competitive
- Takes pride in maintaining high academic standards, even with external content
- Needs to show leadership that investments in new learning tools are paying off

Frustrations:

- The process of approving new courses is painfully slow—too much paperwork and back-and-forth
- She's heard stories of fake certificates and worries about the credibility of students' claimed skills
- Has no easy way to see if students are actually engaging with the extra courses she arranges for them

3. The Course Provider – Alex Rivera

- **Role:** Freelance Developer / Tech Educator

Personal Motivations:

- Wants to turn his years of coding experience into a sustainable income stream
- Hopes to inspire and educate the next generation of developers
- Looking for a way to reach more learners without spending all his time on marketing

Frustrations:

- Setting up his own website for courses is a nightmare—too much tech work, not enough teaching
- Feels invisible trying to compete with giant platforms that already have all the students
- Hates dealing with payment issues, access problems, and the business side of things when he just wants to teach

6. MINIMUM VIABLE PRODUCT (MVP)

The Minimum Viable Product (MVP) of Eduverse represents the first working version of the platform that includes the essential features required to deliver value to users. It allows students, course providers, universities, and administrators to interact with the system and validate the core functionality of the platform.

User Authentication & Role Management:-
Secure registration and login system with Role-Based Access Control (RBAC) that manages different user roles such as Student, Course Provider, University Admin, and Super Admin.

Course Creation & Management:-
Course providers can create and upload courses with details such as title, description, price, lessons, and learning materials like videos and PDFs.

Course Enrollment & Learning Access:-
Students can browse available courses, enroll in them, and access learning materials including lessons, videos, and quizzes.

Quiz & Assessment System:-
Students can attempt quizzes related to their enrolled courses to evaluate their knowledge and track learning progress.

Certificate Generation:-
Students receive a digital certificate after successfully completing course content and assessments.

Admin Dashboard & Platform Management:-
Super Admin can manage users, courses, and universities while monitoring overall platform activity and ensuring smooth system operation.

7. TECHNOLOGIES

React.JS (Frontend)

- Chosen for its component-based architecture, virtual DOM for performance, and extensive ecosystem. Enables building a dynamic, responsive user interface with reusable components across different user roles.

Next.Js, Nest.JS (Backend)

- Selected for its modular architecture, TypeScript support, and dependency injection. Provides excellent structure for scalable server-side applications, ideal for Eduverse's multi-role system.

Express.JS + Node.JS

- Lightweight foundation that works seamlessly with NestJS. Offers non-blocking I/O for handling concurrent user sessions – critical for high-traffic learning platforms.

MongoDB

- Chosen for its flexible document-based data model, which perfectly aligns with Eduverse's varied content structures including user profiles, courses, lessons, and submissions. Offers horizontal scalability, fast read/write operations, and native JSON-like documents that integrate seamlessly with the JavaScript/TypeScript stack. The schema-less design allows for agile development and easy adaptation as platform requirements evolve.

GitHub

- Enables version control, branch management, and collaborative development. Provides clear history of contributions and facilitates code reviews.

8. Algorithms

1. Role-Based Access Control (RBAC) Algorithm

- **Purpose:** Secure user authentication & permission management
- **Logic:** Hierarchical role verification (Super-Admin → University-Admin → Course-Provider → Student)

2. Collaborative Filtering Algorithm

- **Purpose:** Personalized course recommendations
- **Logic:** User-based filtering (students with similar profiles) + Content-based filtering (course attributes)

3. Domain Matching Algorithm

- **Purpose:** University verification & student association
- **Logic:** Email domain extraction + pattern matching against registered university domains

4. Payment Gateway Integration Algorithm

- **Purpose:** Secure transaction processing & subscription management
- **Logic:** Tokenization + encryption + 3D secure authentication

5. Weighted Scoring Algorithm

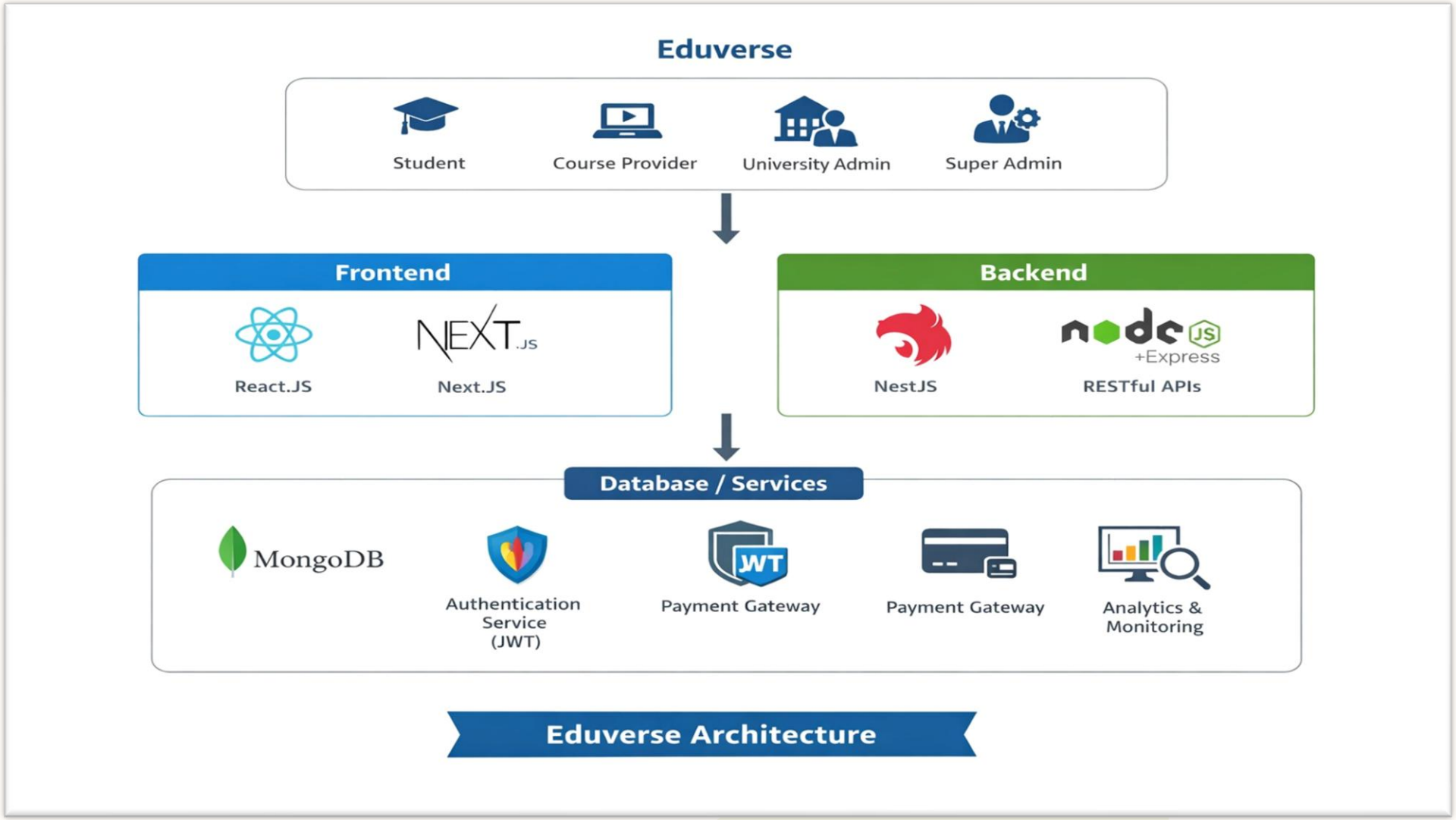
- **Purpose:** Student progress tracking & completion status
- **Logic:** Multi-factor weighted calculation (Lessons 40% + Quizzes 30% + Assignments 20% + Time 10%)

6. Approval Workflow Algorithm

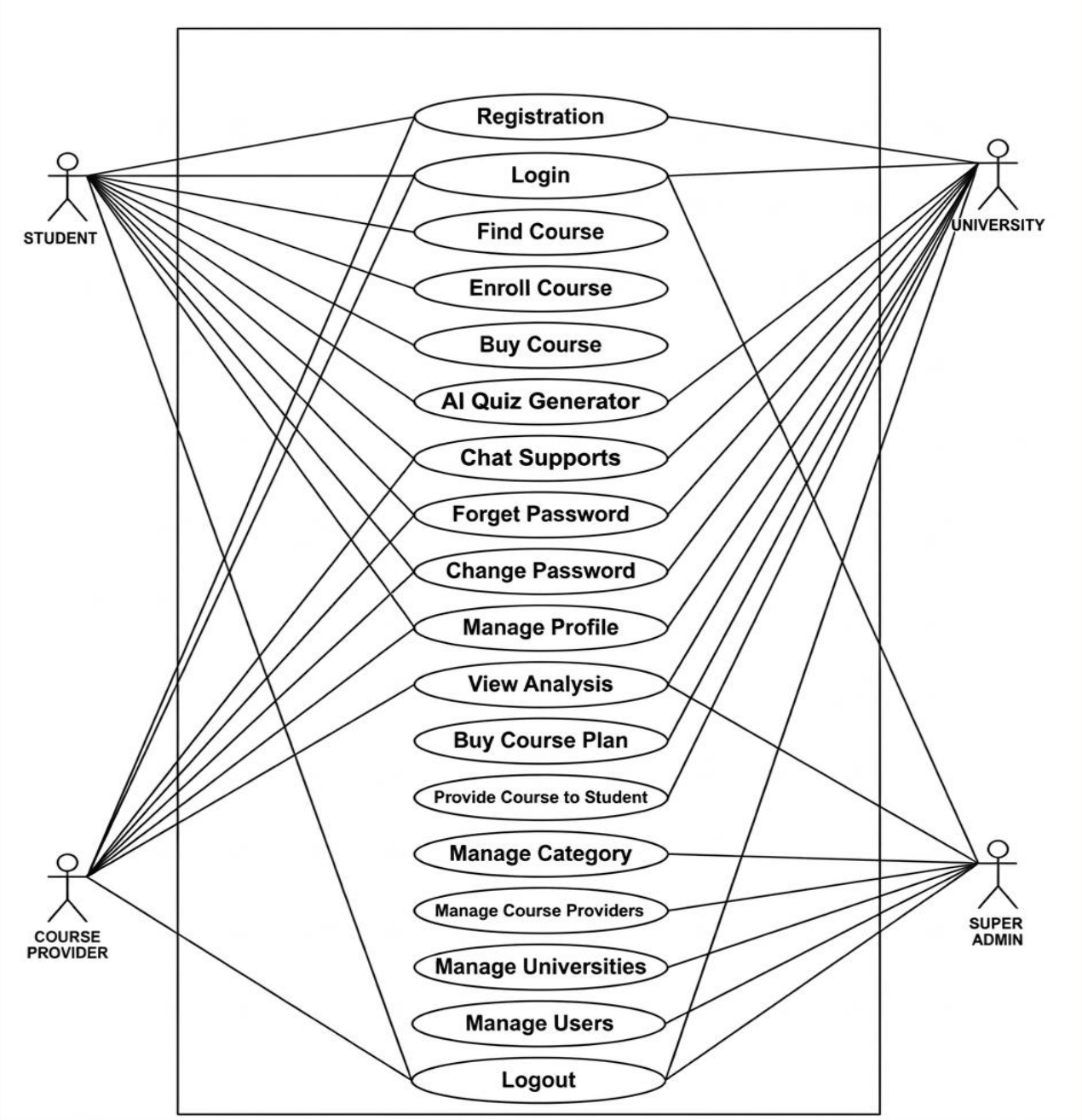
- **Purpose:** Course submission & review process
- **Logic:** State machine (Pending → Review → Approved/Rejected) with notifications

9. DIAGRAMS

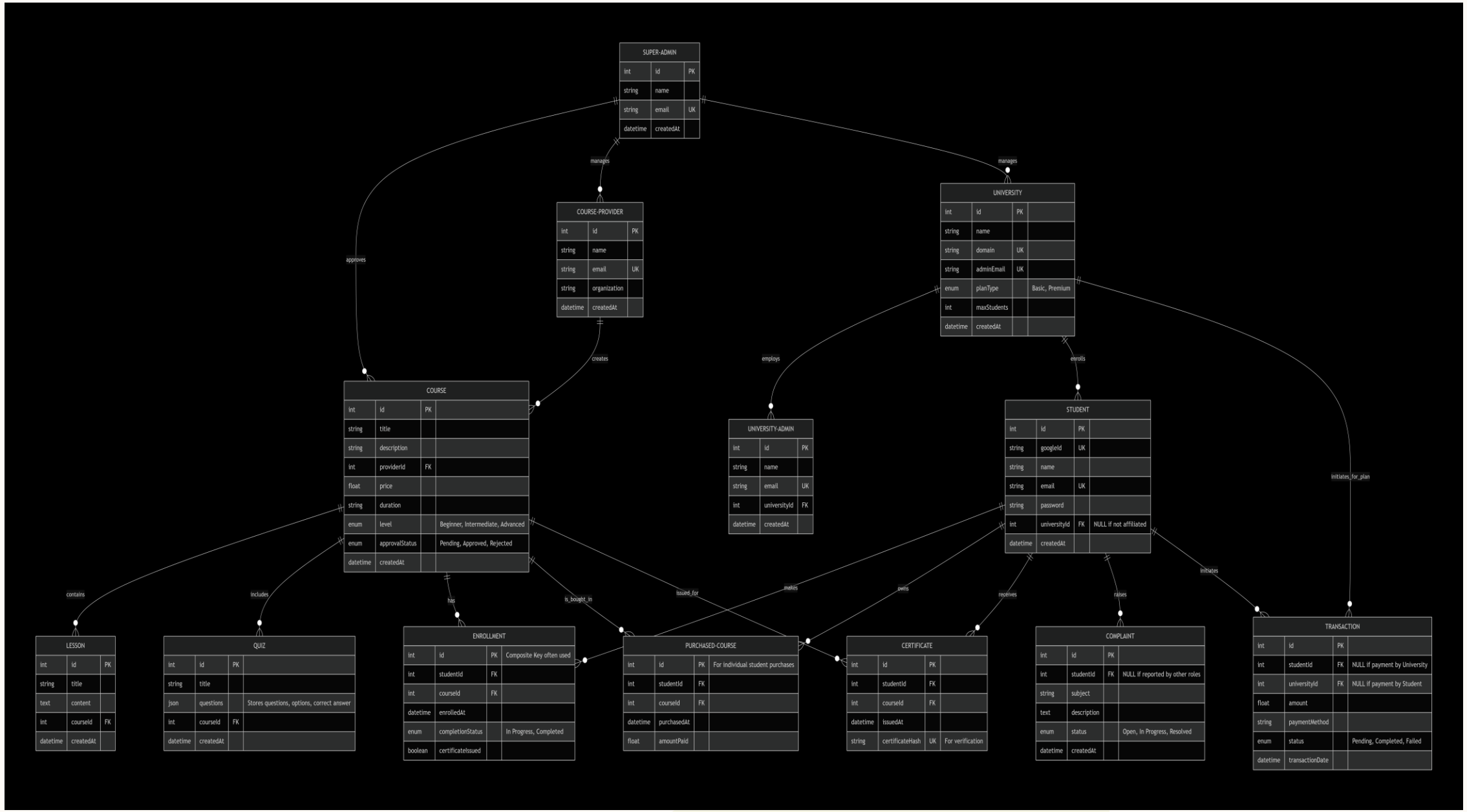
1. Architecture Diagram



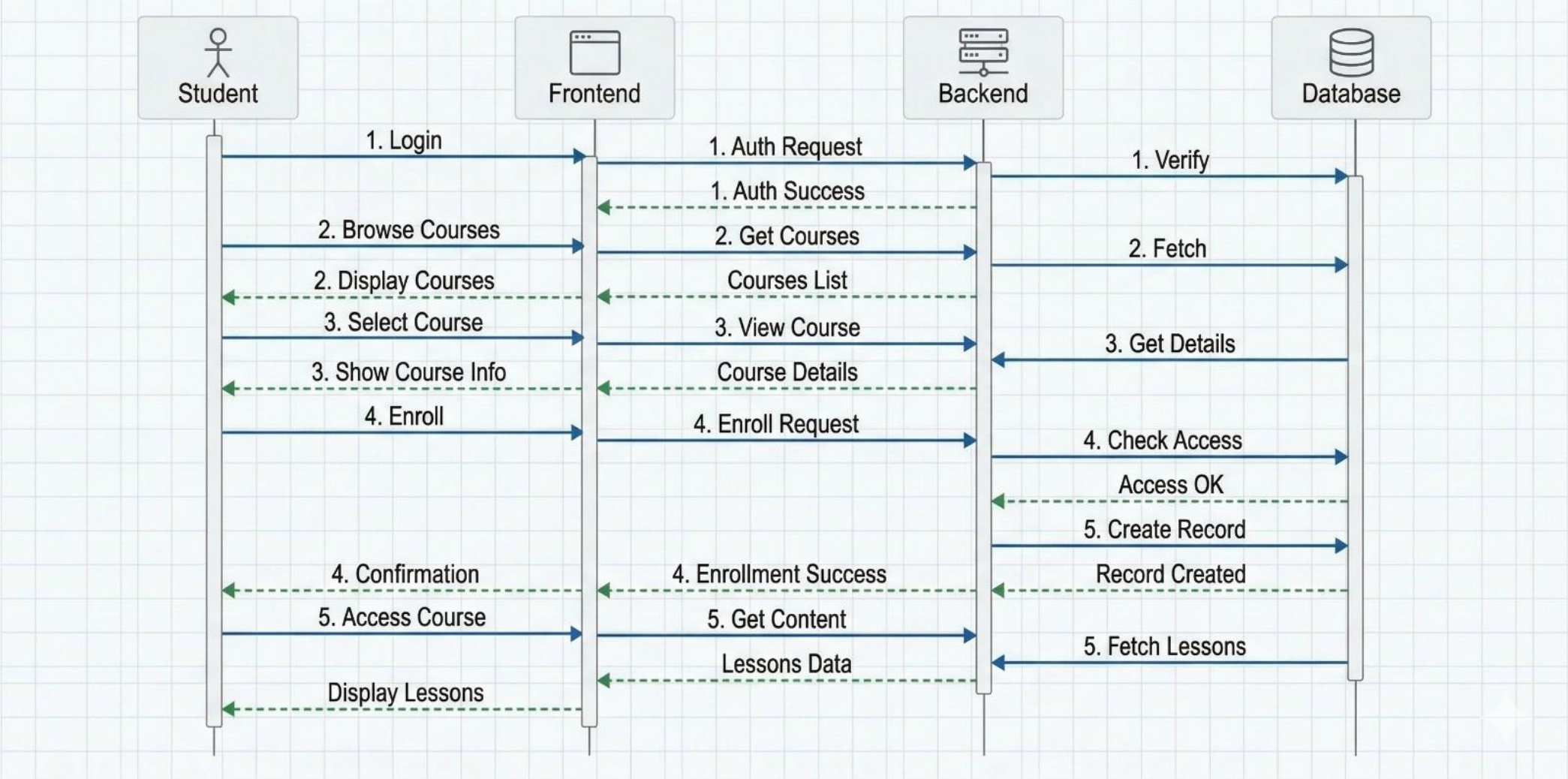
2. Use Case Diagram



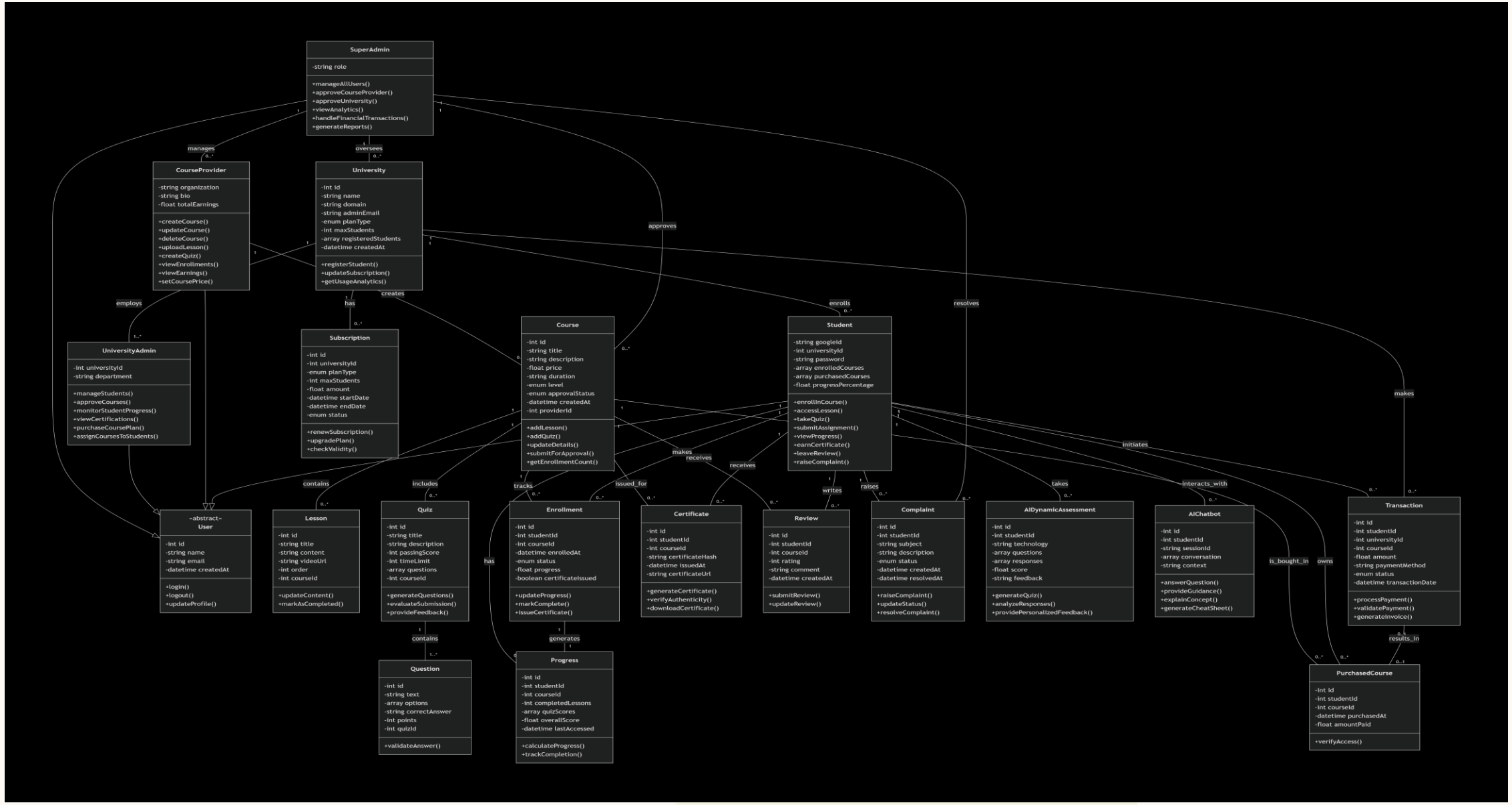
3. ER - Diagram


SUPER-ADMIN
int id PK
string name
string email UK
datetime createdAt
manages
manages
approves
COURSE-PROVIDER
int id PK
string name
string email UK
string organization
datetime createdAt
UNIVERSITY
int id PK
string name
string domain UK
string adminEmail UK
enum planType Basic, Premium
int maxStudents
datetime createdAt
creates
employs
enrolls
COURSE
int id PK
string title
string description
int providerId FK
float price
string duration
enum level Beginner, Intermediate, Advanced
enum approvalStatus Pending, Approved, Rejected
datetime createdAt
UNIVERSITY-ADMIN
int id PK
string name
string email UK
int universityId FK
datetime createdAt
STUDENT
int id PK
string googleId UK
string name
string email UK
string password
int universityId FK NULL if not affiliated
datetime createdAt
initiates_for_plan
contains
includes
has
is_bought_in
issued_for
makes
owns
receives
raises
initiates
LESSON
int id PK
string title
text content
int courseId FK
datetime createdAt
QUIZ
int id PK
string title
json questions Stores questions, options, correct answer
int courseId FK
datetime createdAt
ENROLLMENT
int id PK Composite Key often used
int studentId FK
int courseId FK
datetime enrolledAt
enum completionStatus In Progress, Completed
boolean certificateIssued
PURCHASED-COURSE
int id PK For individual student purchases
int studentId FK
int courseId FK
datetime purchasedAt
float amountPaid
CERTIFICATE
int id PK
int studentId FK
int courseId FK
datetime issuedAt
string certificateHash UK For verification
COMPLAINT
int id PK
int studentId FK NULL if reported by other roles
string subject
text description
enum status Open, In Progress, Resolved
datetime createdAt
TRANSACTION
int id PK
int studentId FK NULL if payment by University
int universityId FK NULL if payment by Student
float amount
string paymentMethod
enum status Pending, Completed, Failed
datetime transactionDate

4. Sequence Diagram - Course Enrollment



5. Class Diagram



10. PRODUCT BACKLOG

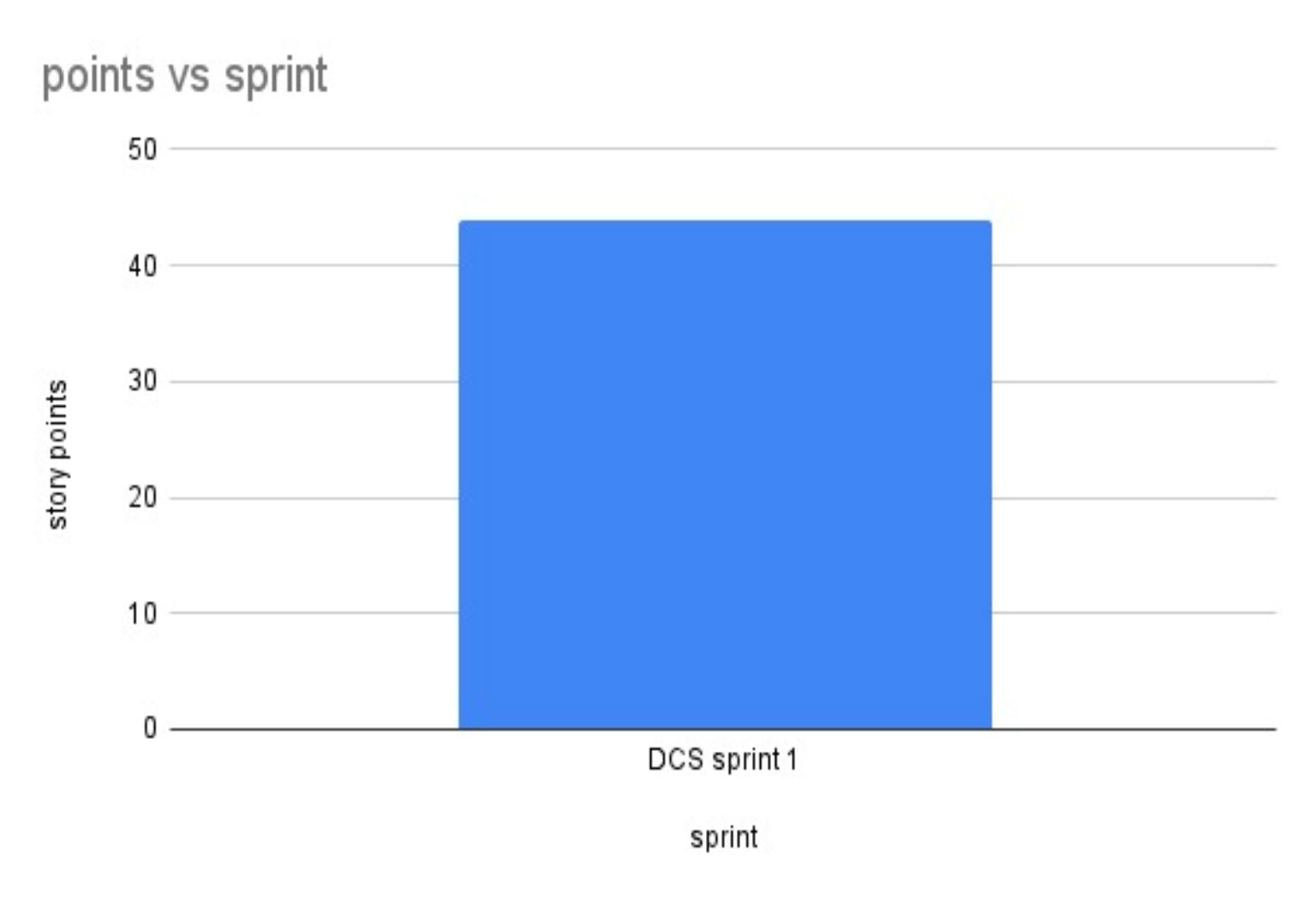
Module	ID	User Story	Story Points	Priority
Project Foundation	PF-01	As a team, we need to define the project objective, scope, and core components.	2	High
Project Foundation	PF-02	As a team, we need to learn and finalize the technology stack (ReactJS, NestJS, PostgreSQL).	10	High
Project Foundation	PF-03	As a developer, I need to design and set up the database schema.	2	High
Project Foundation	PF-04	As a developer, I need to define the core modules of the application (User, Course, University, etc.).	2	High
Model Design	MD-01	As a developer, I need to develop the core data models for the application.	5	High
Model Design	MD-02	As a developer, I need to test the data models for integrity and performance.	5	Medium
User Interface	UI-01	As a UI/UX designer, I need to design a user-friendly and responsive interface for all user roles.	10	High
Authentication & Authorization	AA-01	As a user, I want to be able to register for an account with my role (Student, University, Provider).	5	High
Authentication & Authorization	AA-02	As a registered user, I want to be able to log in securely to the platform.	5	High
Authentication & Authorization	AA-03	As a user, I want to manage my profile, update my details, and change my password.	5	Medium
Course Management	CM-01	As a course provider, I want to create new courses and upload multimedia content (videos, PDFs).	10	High
Course Management	CM-02	As a course provider, I want to set prices and manage the availability of my courses.	8	High
Course Management	CM-03	As a course provider, I want to view and manage student enrollments for my courses.	3	Medium
Course Discovery	CD-01	As a student, I want to search for and browse available courses.	3	High
Course Discovery	CD-02	As a student, I want to view detailed information about a course before enrolling.	3	High
Learning Experience	LE-01	As a student, I want to enroll in a course and access all its lessons and materials.	5	High
Learning Experience	LE-02	As a student, I want to take quizzes and complete assessments within a course.	5	High
Learning Experience	LE-03	New: AI Quiz Generator As a course provider, I want to automatically generate quizzes based on my course content using AI to save time and create assessments quickly.	8	Medium
Learning Experience	LE-04	As a student, I want to receive a certificate upon successfully completing a course.	2	High
Learning Experience	LE-05	As a student, I want to provide feedback and rate the courses I have completed.	5	Low
University Management	UM-01	As a university admin, I want to browse and purchase course plans for my institution.	5	High
University Management	UM-02	As a university admin, I want to assign purchased course plans to specific students.	5	High
AI-Powered Features	AI-01	New: AI Chatbot Assistant As a student, I want to interact with an AI-powered chatbot to get instant answers to my questions about courses, navigation, or platform features.	8	Medium
Super Admin	SA-01	As a super admin, I want to manage all users (students, universities, providers) on the platform.	5	High
Super Admin	SA-02	As a super admin, I need to approve new course providers and university registrations.	5	High
Super Admin	SA-03	As a super admin, I want to view detailed analytics on platform usage, revenue, and course performance.	5	Medium
Payments	PAY-01	As a user, I want to securely purchase courses or university plans using a payment gateway.	8	High
Provider Analytics	PA-01	As a course provider, I want to analyze student engagement, sales, and course ratings.	5	Medium
System & Compliance	SYS-01	As a developer, I need to ensure the platform adheres to data protection policies and educational standards.	5	High

11. SPRINT 1 BACKLOG

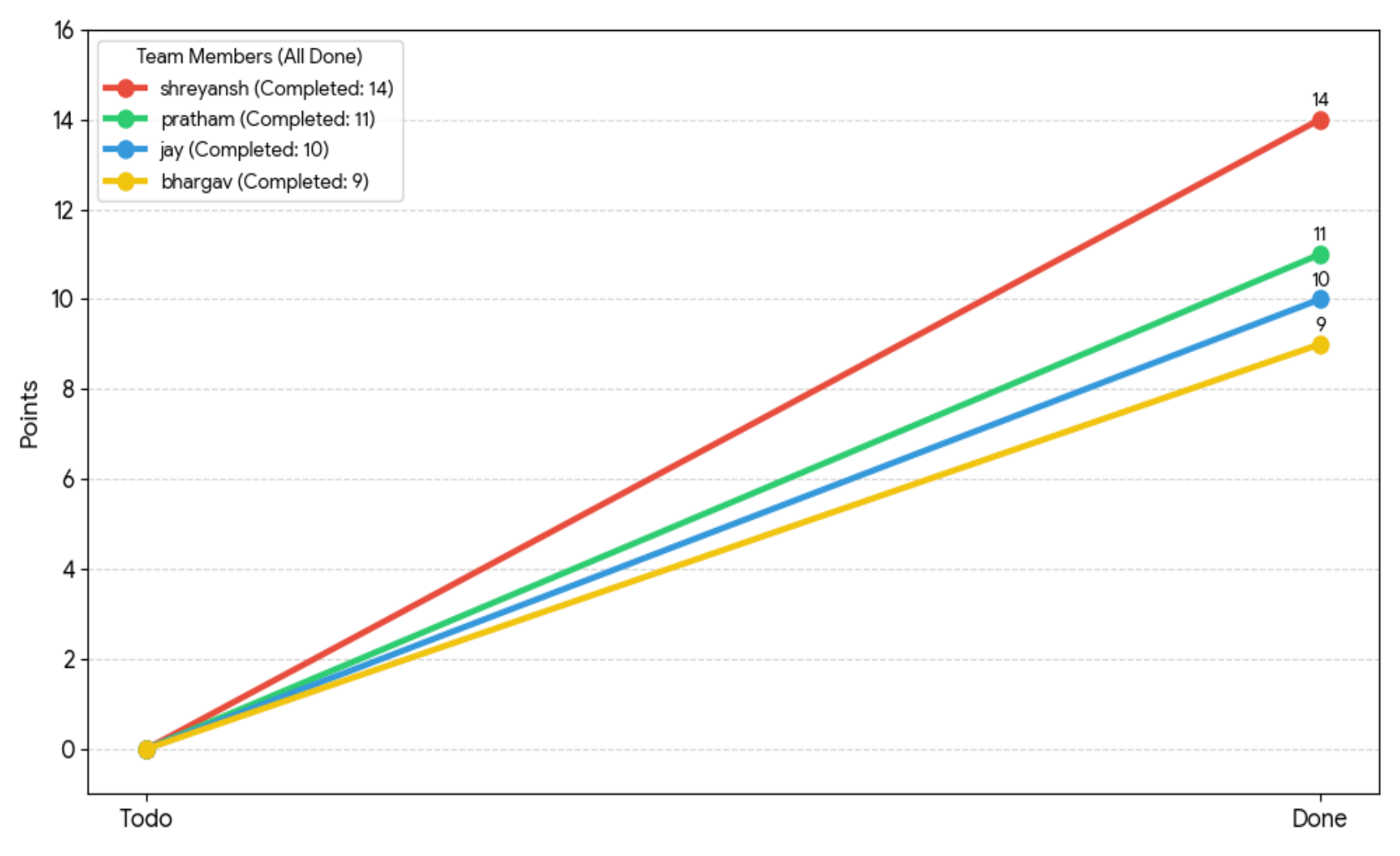
Task ID	Module Description	User Story	Story Point	Priority
AA-01	Authentication & Authorization	As a user, I want to be able to register for an account with my role (Student, University, Course Provider).	5	High
AA-02	Authentication & Authorization	As a registered user, I want to be able to log in securely to the platform.	5	High
AA-03	Authentication & Authorization	As a user, I want to manage my profile, update my details, and change my password.	5	Medium
CM-01	Course Management	As a course provider, I want to create new courses and upload multimedia content (videos, PDFs).	10	High
CM-02	Course Management	As a course provider, I want to set prices and manage the availability of my courses.	8	High
CD-01	Course Discovery	As a student, I want to search for and browse available courses.	3	High
CD-02	Course Discovery	As a student, I want to view detailed information about a course before enrolling.	3	High
LE-01	Learning Experience	As a student, I want to enroll in a course and access all its lessons and materials.	5	High

12. METRICS

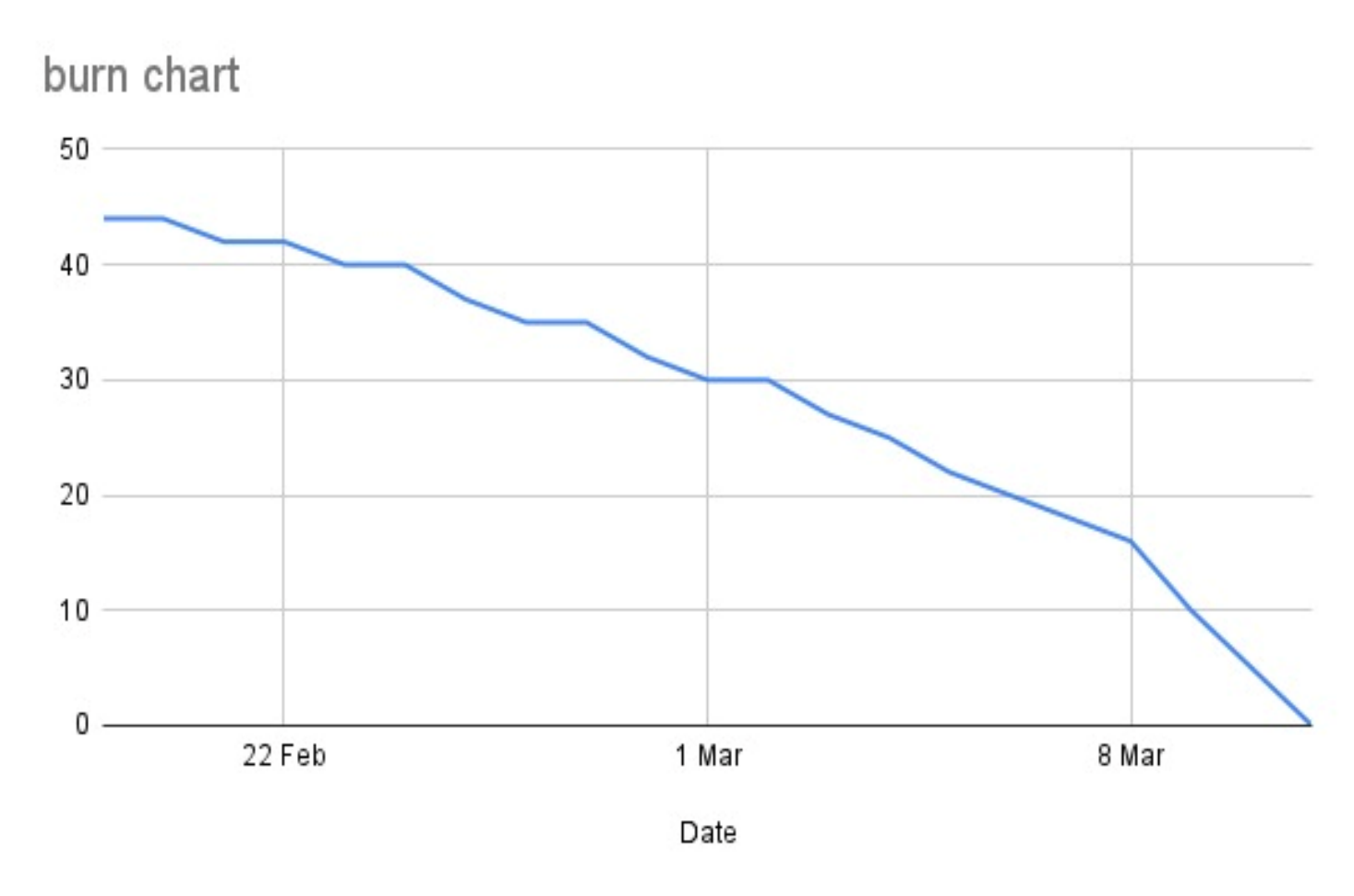
- Metrics Velocity Report



- **Team Members Velocity Report**



- **Sprint Burndown Chart**



13. RETROSPECTIVE

What Went Well:

- **Effective Agile Adaptation:** We successfully transitioned from planning to execution. The team demonstrated its agility by taking the user stories, personas, and technical design from Sprint 0 and turning them into functional code. The focus on the MVP features kept development aligned with our core objectives.

- **Solid Technical Foundation:** The team made significant progress on the platform's backbone. The work on implementing the core algorithms (like RBAC and Domain Matching) and setting up the architecture (as per our diagrams) has created a stable and scalable foundation for future sprints.

- **Clear Progress on MVP Features:** We successfully began development on key MVP components. The progress on user authentication, role management, and the initial course creation workflows shows a tangible move towards a working product.

What Needs Improvement:

- **Proactive Dependency Management:** This sprint revealed some dependencies between tasks (e.g., Course Enrollment depending on a stable Authentication system). While we managed them, we can improve by identifying and communicating these dependencies earlier in the sprint to prevent blocking team members.

- **Integrating Continuous Learning:** The goal from the last sprint to focus on continuous learning was partially met through self-learning during development. However, we can improve by dedicating specific time for knowledge sharing, such as a brief "tech talk" on a new tool or a complex part of the codebase we mastered during the sprint.

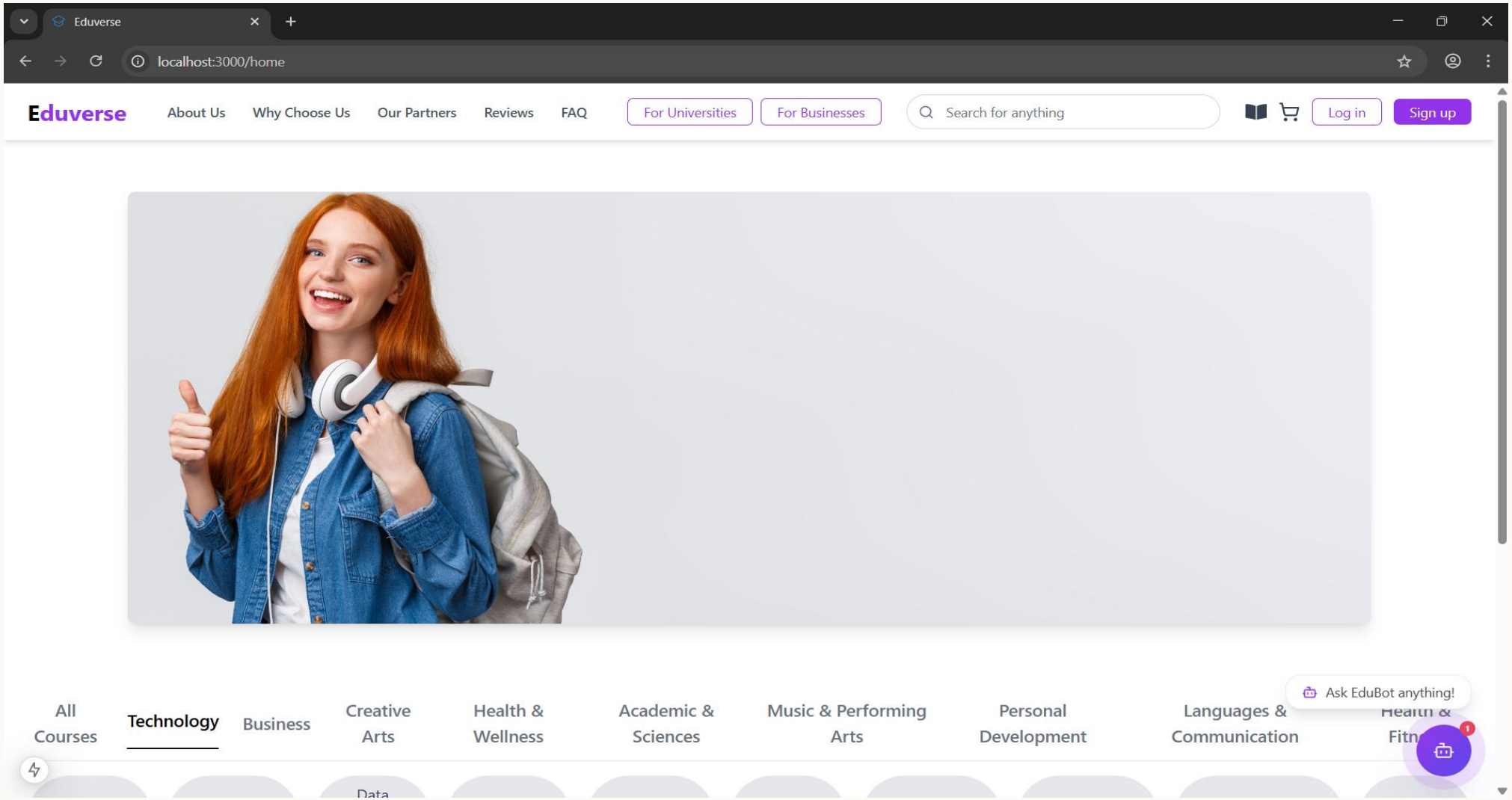
Goal For Next Sprint:

- Complete the core learning cycle by integrating the Quiz & Assessment System with Certificate Generation, ensuring students can seamlessly transition from learning to assessment to certification. Additionally, we will focus on enhancing the Admin Dashboard to provide meaningful insights into platform activity and user engagement."

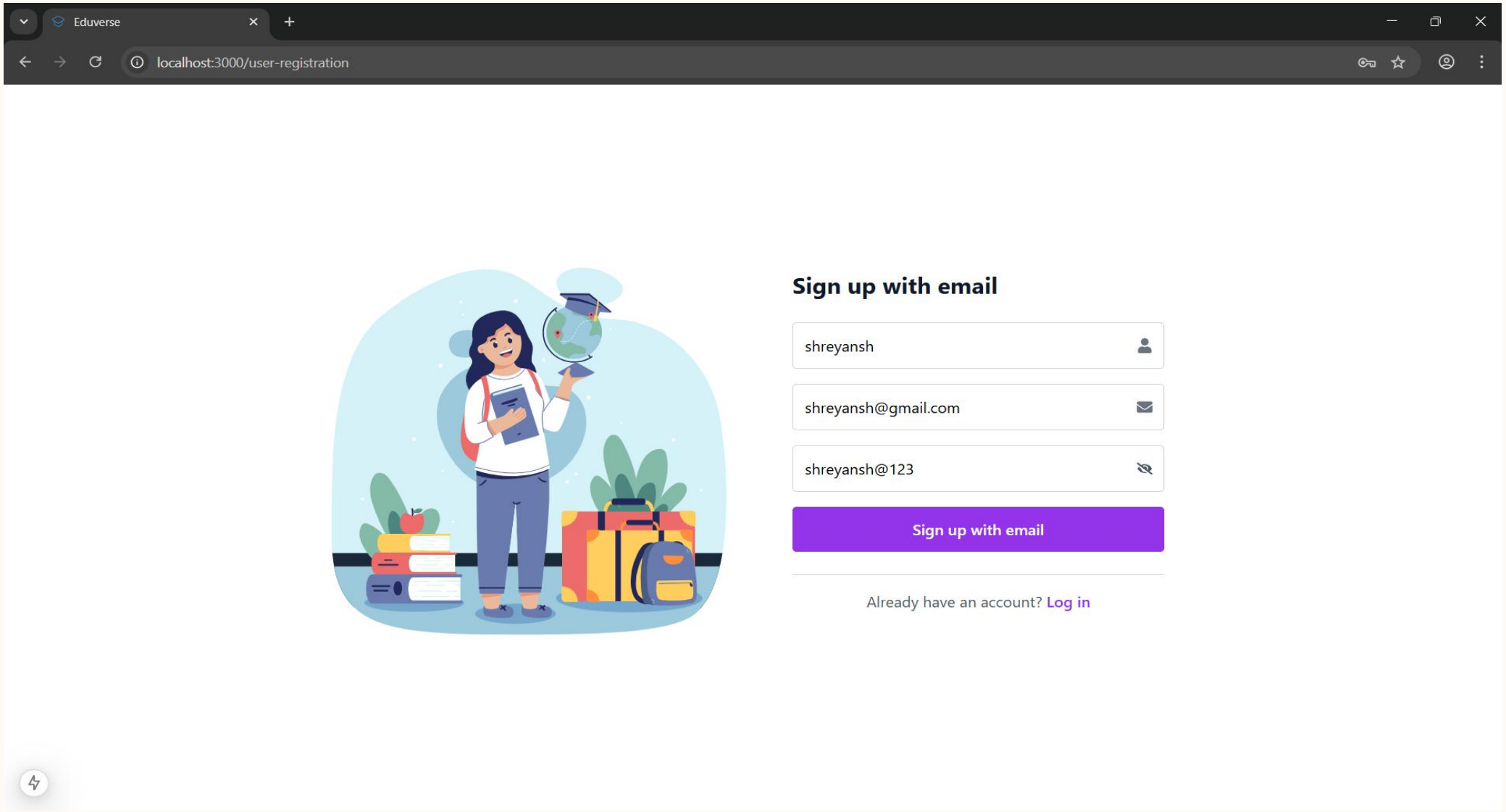
14. SPRINT 2 PLANNED STORIES

Task ID	Module Description	User Story	Story Point	Priority
LE-02	Learning Experience	As a student, I want to take quizzes and complete assessments within a course.	5	High
LE-04	Learning Experience	As a student, I want to receive a certificate upon successfully completing a course..	5	High
LE-05	Learning Experience	As a student, I want to provide feedback and rate the courses I have completed.	5	Medium
UM-01	University Management	As a university admin, I want to browse and purchase course plans for my institution.	10	High
UM-02	University Management	As a university admin, I want to assign purchased course plans to specific students.	8	High
CM-03	Course Management	As a course provider, I want to view and manage student enrollments for my courses.	3	High
PAY-01	Payments	As a user, I want to securely purchase courses or university plans using a payment gateway.	3	High

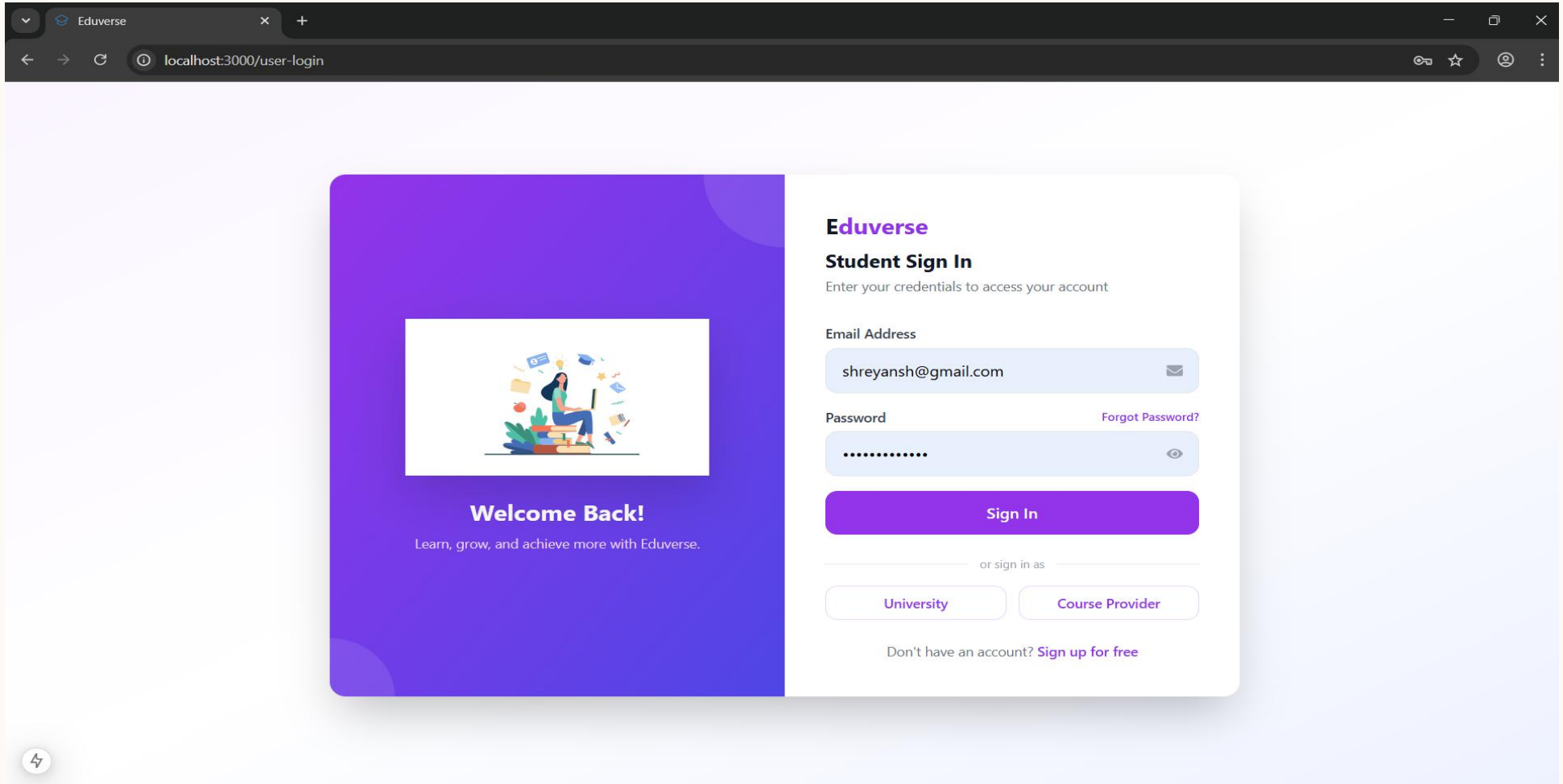
15. PROJECT DEMO - SPRINT 1



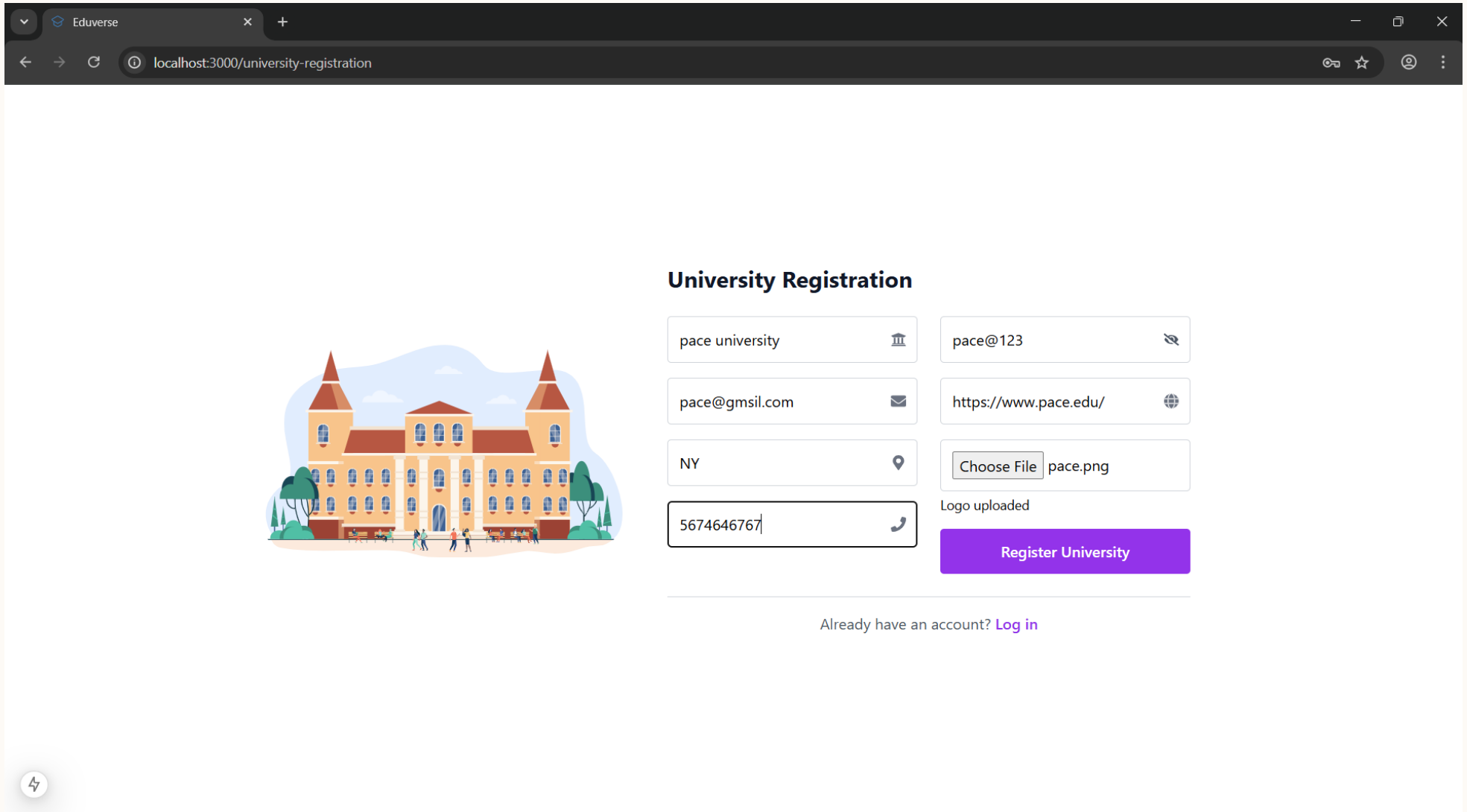
1. User_HomePage



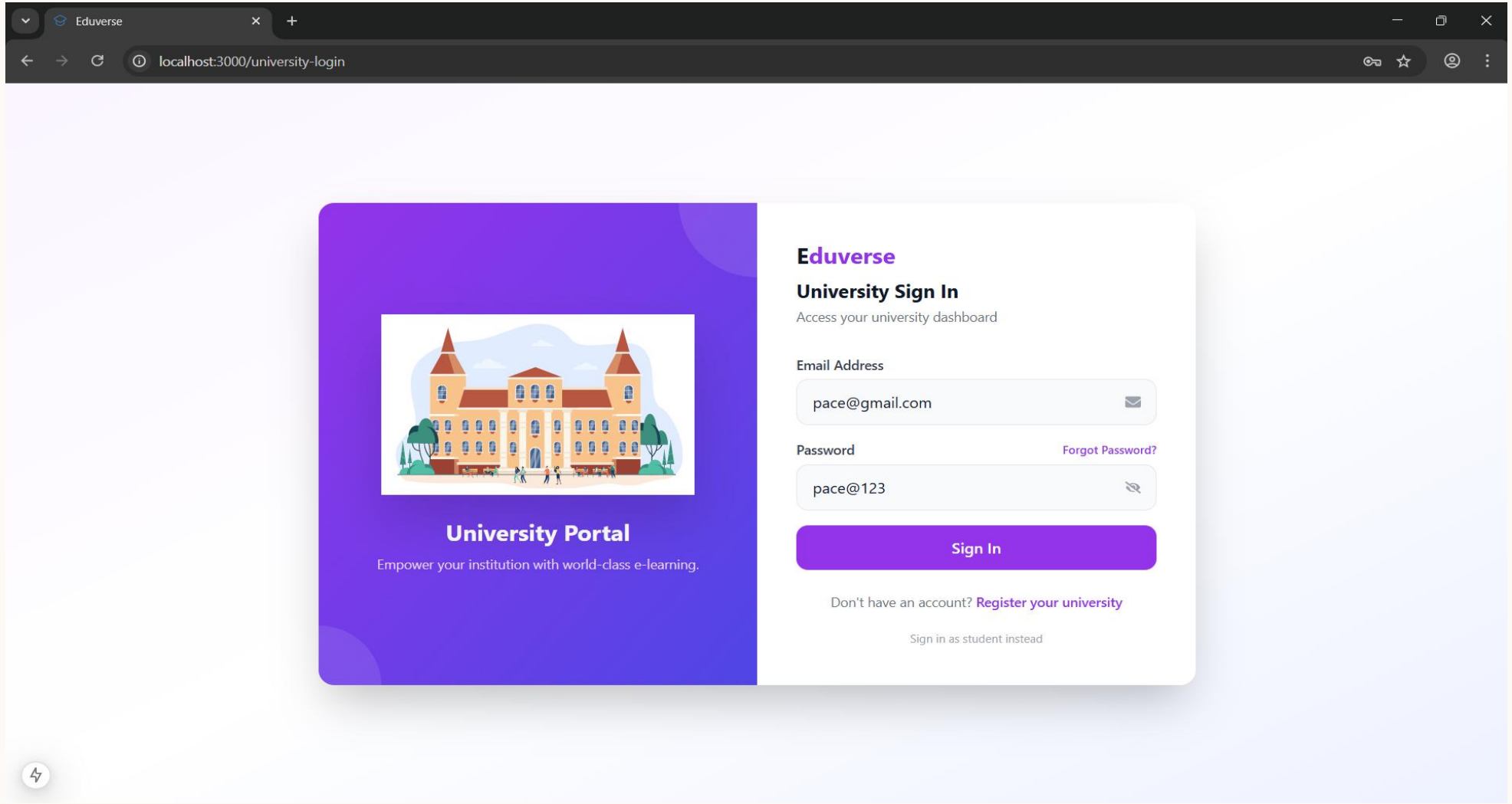
2. User_Registration


Eduverse
localhost:3000/user-login
Welcome Back!
Learn, grow, and achieve more with Eduverse.
Eduverse
Student Sign In
Enter your credentials to access your account
Email Address
shreyansh@gmail.com
Password
Forgot Password?
Sign In
or sign in as
University
Course Provider
Don't have an account? Sign up for free

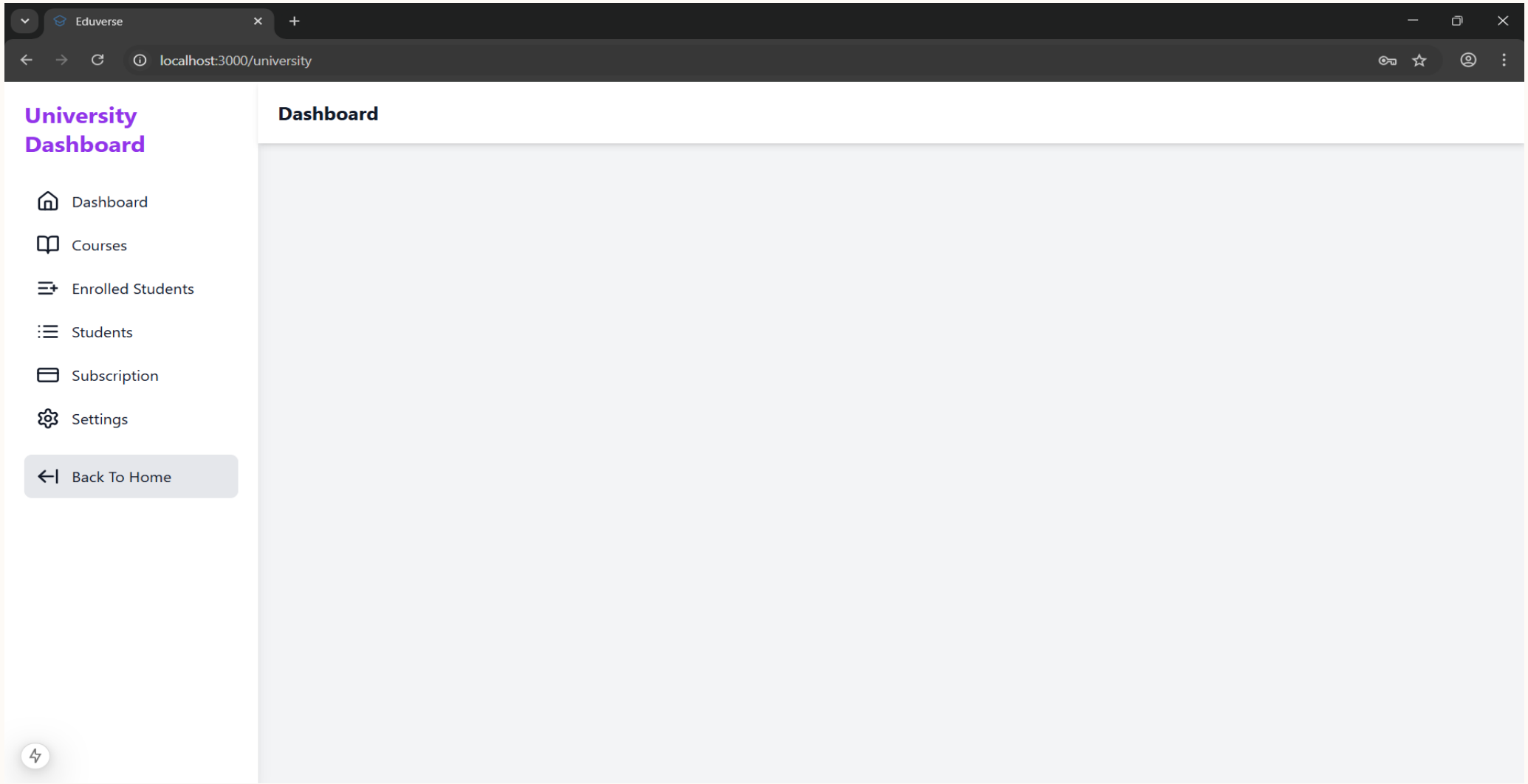
3. User Login



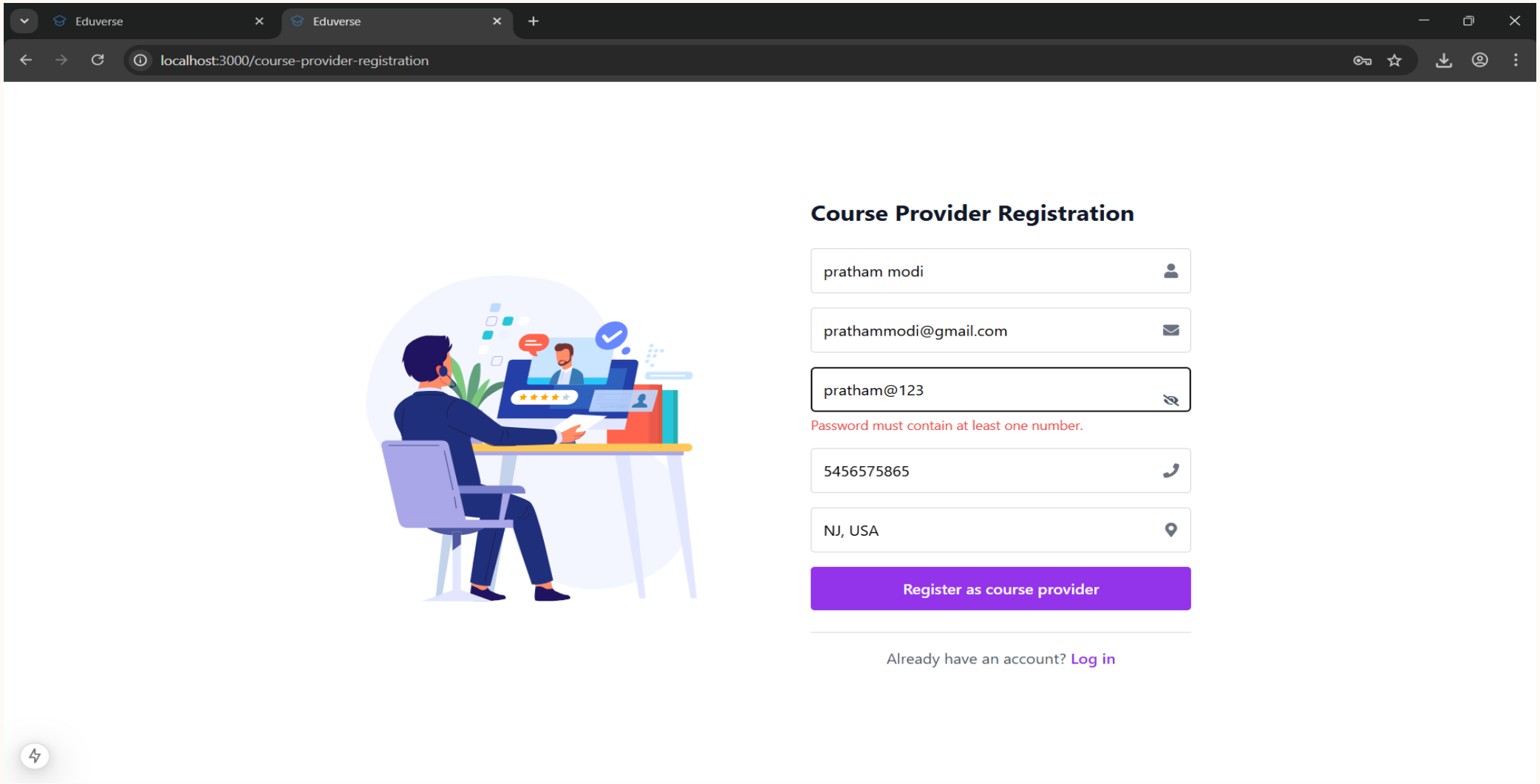
4. University Registration



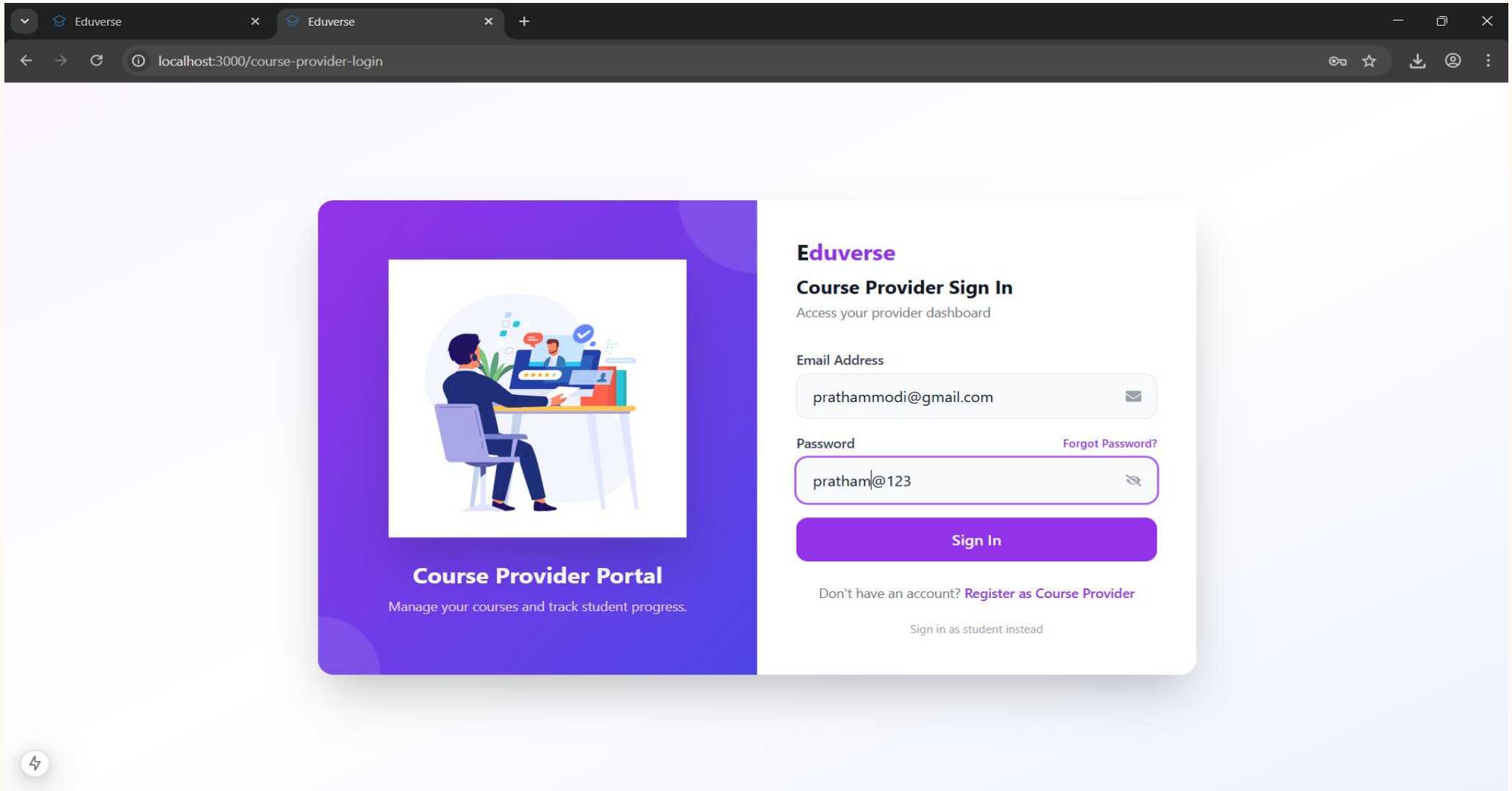
5. University login



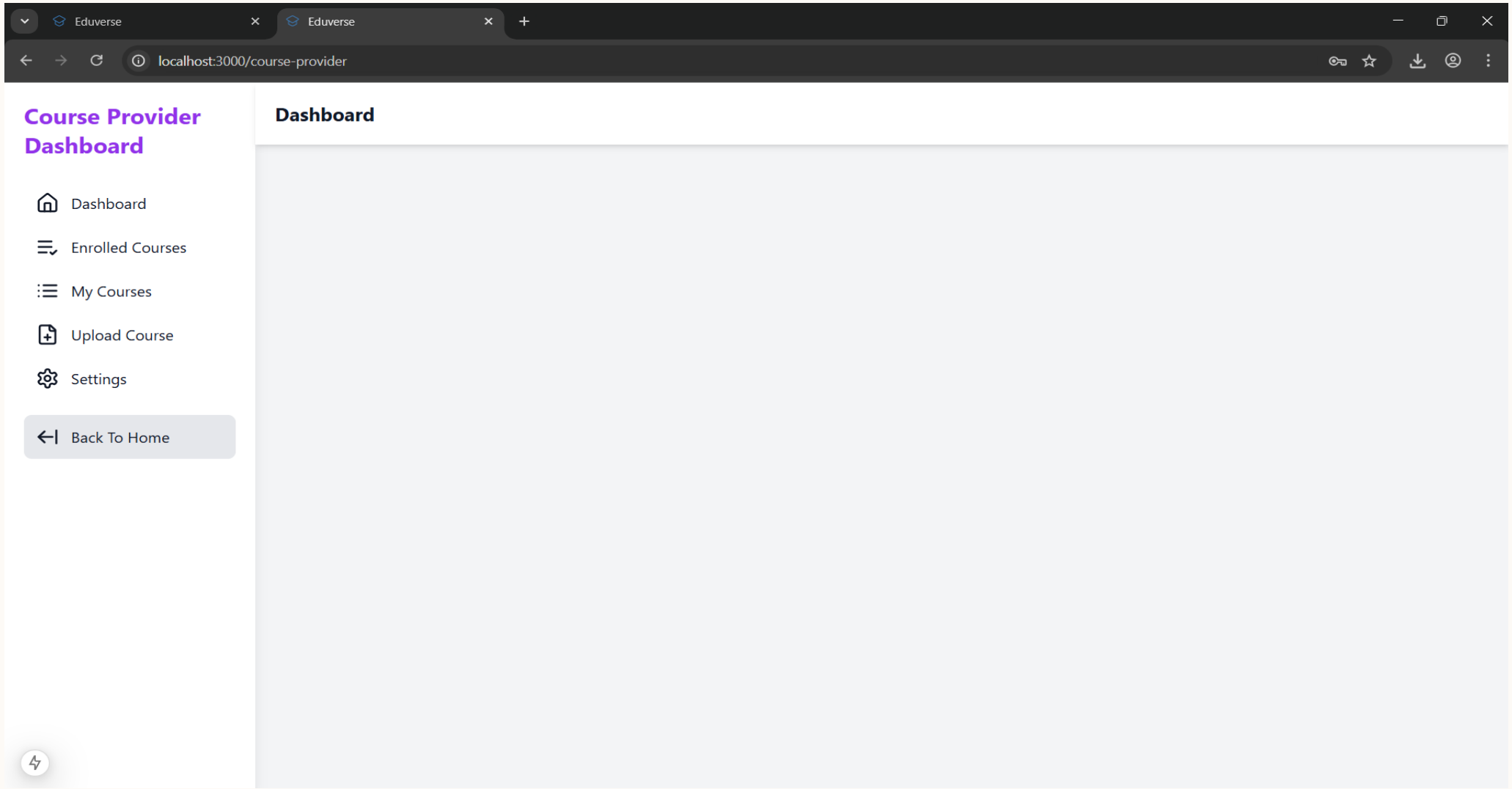
6. University Dashboard



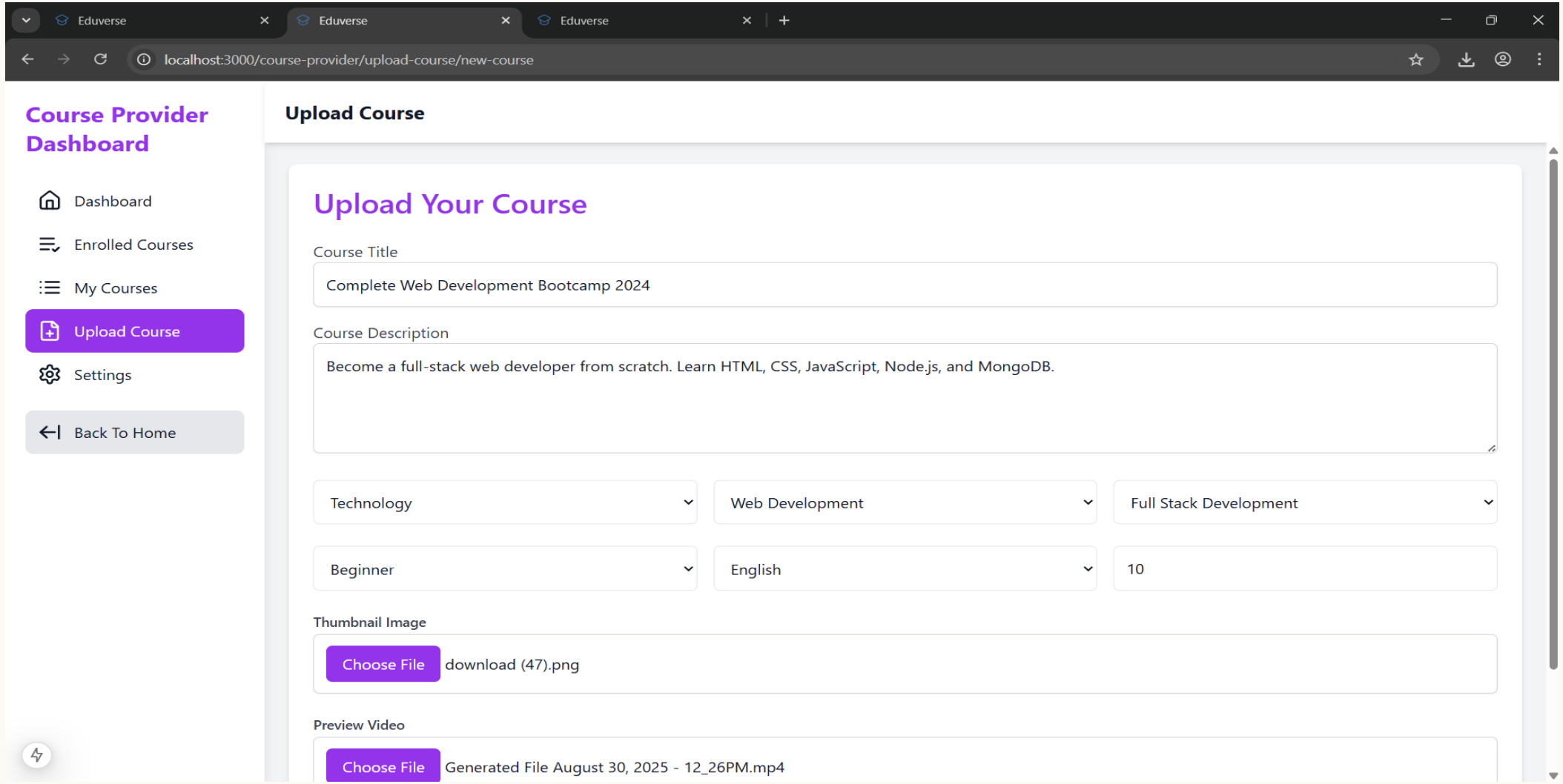
7. Course Provide Registration



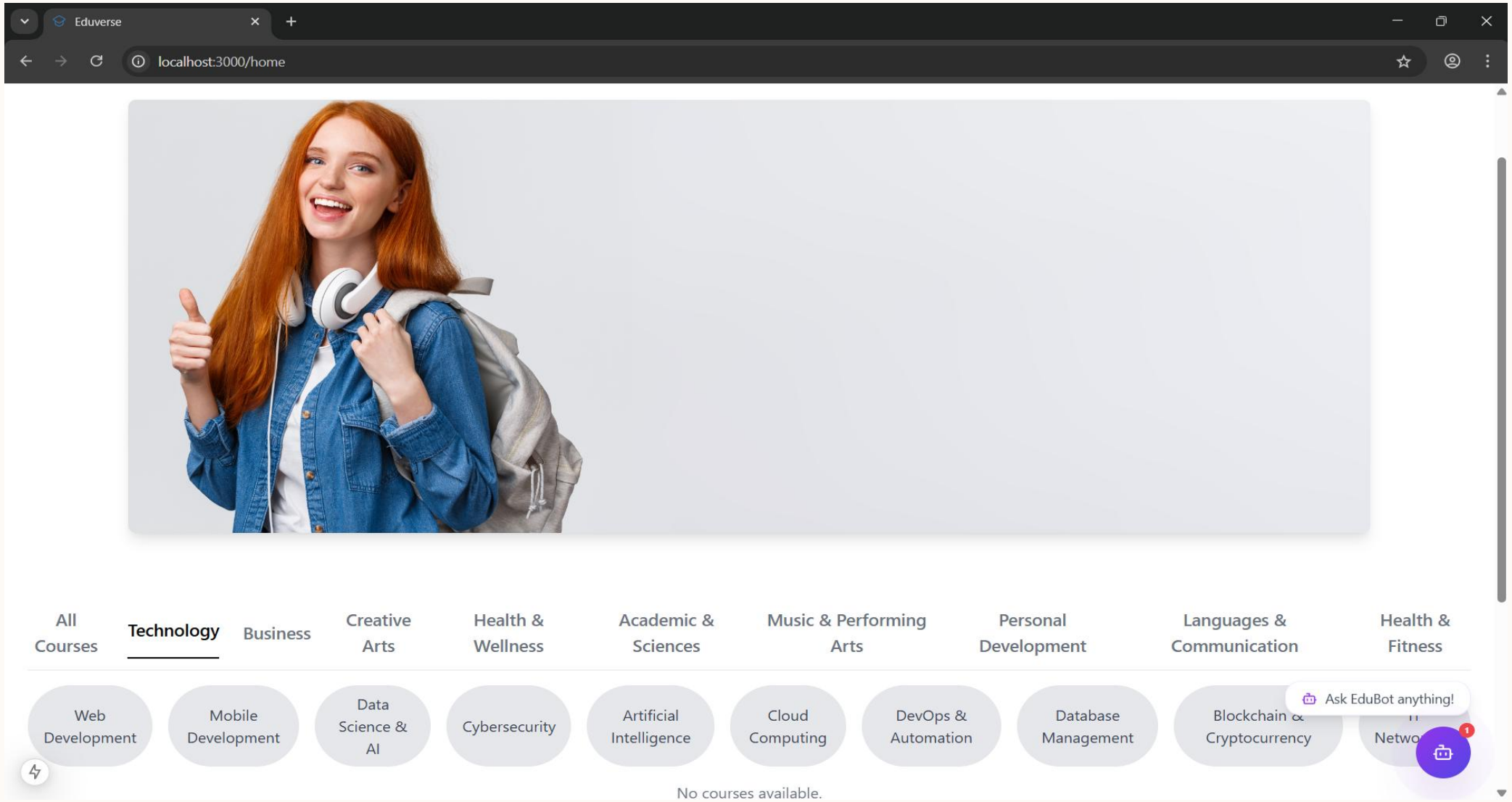
9. Course Provider Login



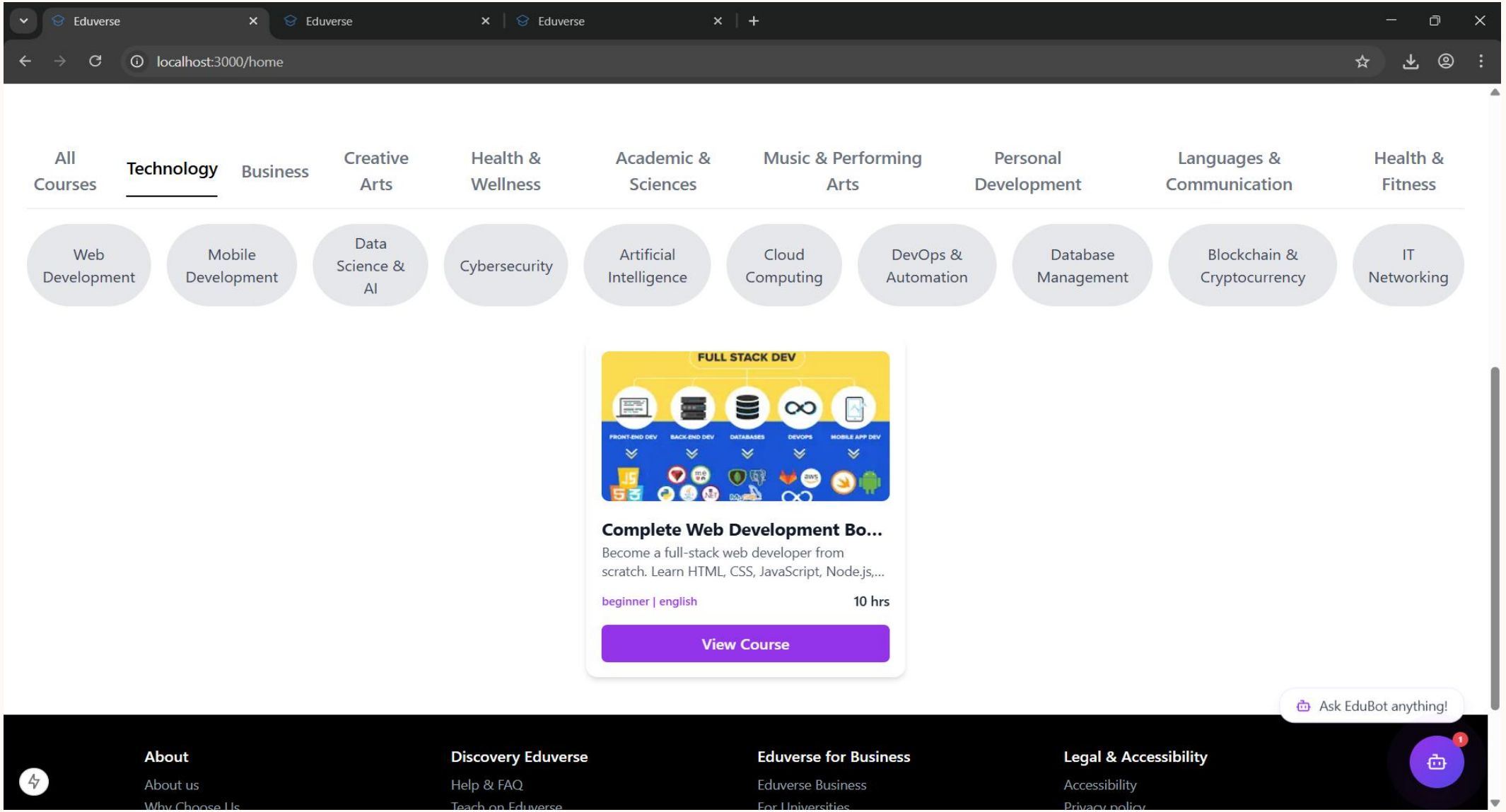
10. Course Provider DashBoard

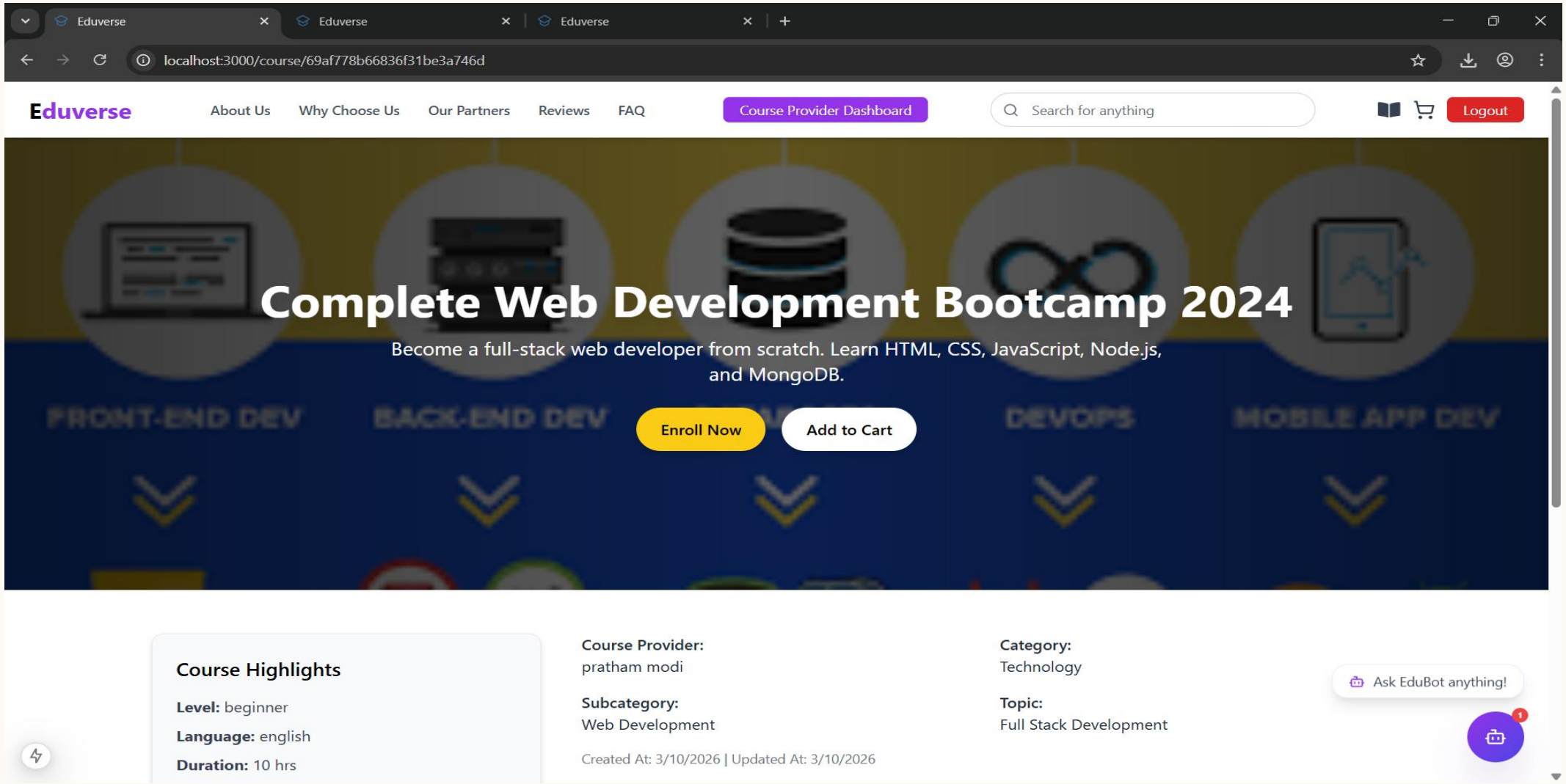


11. Course Provider – Upload Course



12. User - Course Dynamic


Eduverse
Eduverse
Eduverse
localhost:3000/home
All Courses
Technology
Business
Creative Arts
Health & Wellness
Academic & Sciences
Music & Performing Arts
Personal Development
Languages & Communication
Health & Fitness
Web Development
Mobile Development
Data Science & AI
Cybersecurity
Artificial Intelligence
Cloud Computing
DevOps & Automation
Database Management
Blockchain & Cryptocurrency
IT Networking
FULL STACK DEV
Complete Web Development Bo...
Become a full-stack web developer from scratch. Learn HTML, CSS, JavaScript, Node.js,...
beginner | english
10 hrs
View Course
Ask EduBot anything!
About
About us
Discovery Eduverse
Help & FAQ
Eduverse for Business
Eduverse Business
Legal & Accessibility
Accessibility



12. User – Enroll Course

16. GITHUB LINK

https://github.com/shreyanshdhanani/eduverse/wiki

THANK YOU

Shreyansh Dhanani
Pratham Modi
Bhargav Kakadiya
Jay Kakadiya